PREDIXION AI | ENGINEERING ASSIGNMENT

Dialer Campaign Analytics

From call logs to actionable collections decisions

Assignment 2 • Part B • Data analysis & recommendations

20,000
accounts

70,235
call records

4
analysis views

Synthetic dataset • seed 42

Four questions, one operational objective

Measure where contact is more likely, where promises convert, and where data quality breaks.

OVERALL CONNECTION RATE

25.7%

Connected dispositions ÷ all call attempts.

WEEKEND CONNECTION RATE

29.3%

vs 23.8% on weekdays.

BEST OBSERVED HOUR

18:00

33.6% connection rate in this dataset.

B7 + BR WRONG-NUMBER

53.5%

213 connected calls in the segment.

01 CONTACT

Hourly and weekday/weekend connection rates

02 CONVERSION

PTP ÷ RPC by DPD; kept PTP by promise horizon

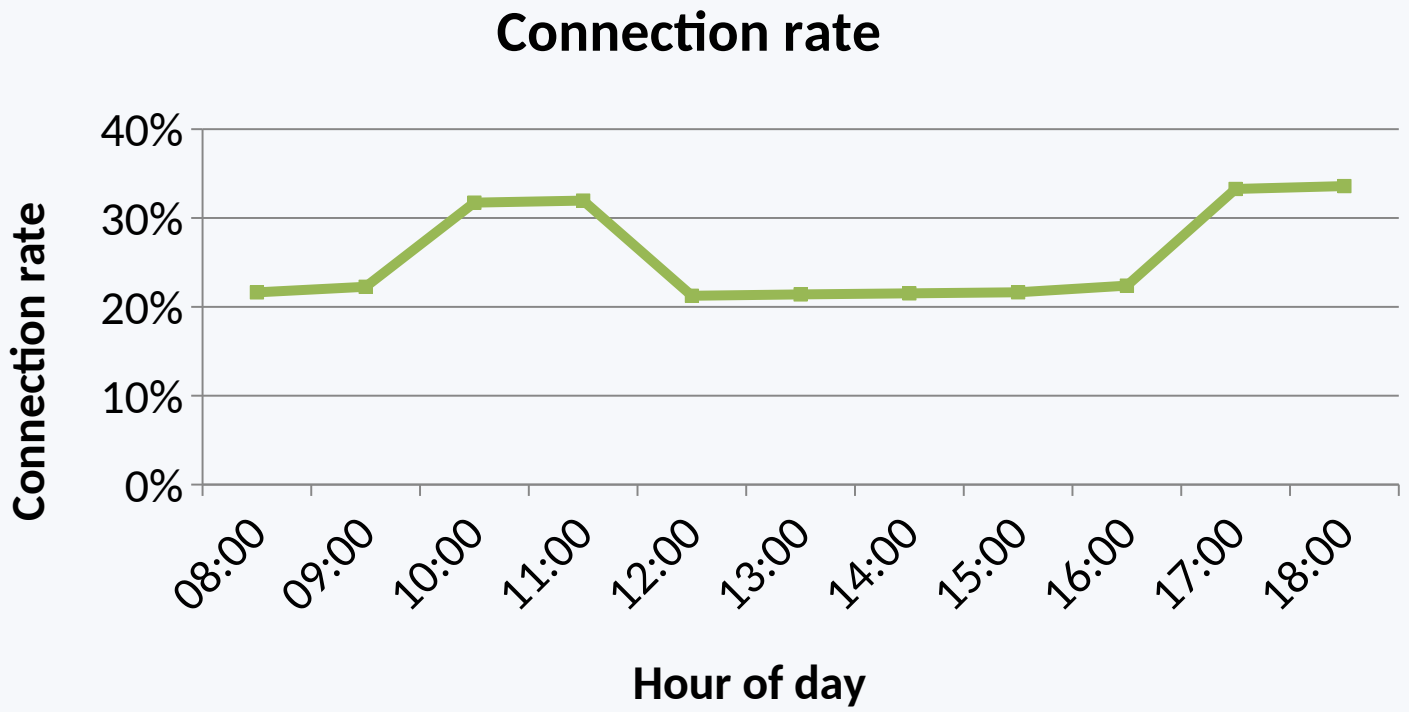
03 DATA QUALITY

State × batch anomaly and targeted action

Decision principle: optimise for verified right-party contact and kept promises—not raw dial volume alone.

Contact rates peak late morning and after 17:00

Connection rate = connected calls ÷ all attempts; each call record is one attempt.



10:00–11:00

31.7–32.0%

Higher than the 21–22% midday baseline.

17:00–18:00

33.3–33.6%

Highest observed time window.

WEEKEND VS WEEKDAY

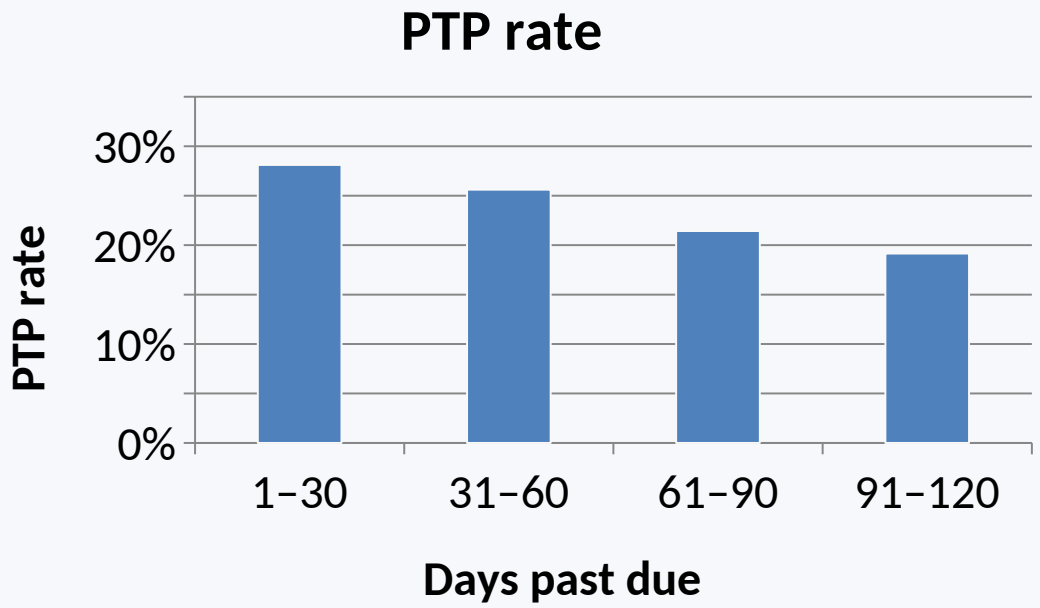
+5.4 pp

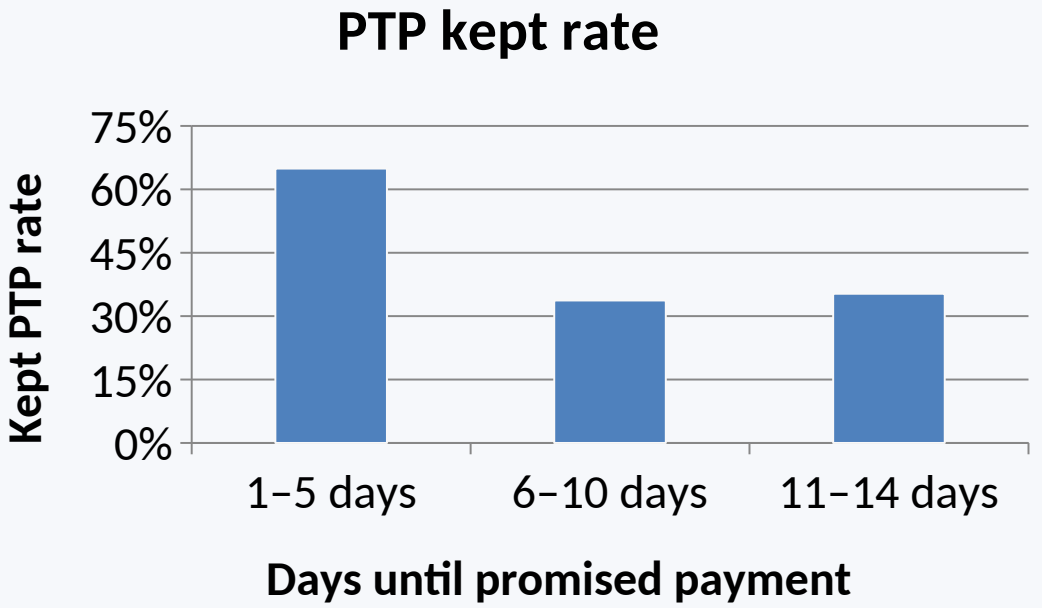
29.3% vs 23.8%.

Action: prioritise controlled calling windows; validate by portfolio and compliance constraints before scaling.

Lower DPD segments show higher PTP rates; short promises are kept more often

PTP rate uses RPC as the denominator. Kept rate uses PTP records only.





PTP rate falls from 28.1% (1–30 DPD) to 19.2% (91–120 DPD).

Promises within 1–5 days are kept at 64.9%, versus ~35% for longer horizons.

B7 in Bihar is a clear data-quality outlier

Wrong-number rate among connected calls; segment is state = BR and batch = B7.

BR / B7 WRONG-NUMBER RATE

53.5%

213 connected calls; 114 wrong numbers.

OTHER SEGMENTS BASELINE

7.8%

Wrong-number rate outside BR / B7.

ESTIMATED EXCESS WRONGS

~97

At the observed segment volume, versus the rest-of-data baseline.

LIKELY ROOT CAUSE

List quality

Investigate stale/mis-mapped phone records and batch ingestion.

Recommended actions

1. **Quarantine & verify** — Pause or reduce BR/B7 dialling; validate a sample against source records before reuse.
2. **Reconcile upstream mapping** — Audit batch-to-state assignment, phone-number freshness, and ingestion/normalisation rules.
3. **Add a monitoring guardrail** — Alert when wrong-number rate exceeds the portfolio baseline by a defined margin; review daily.

Run a controlled optimisation—not a blanket dialer change

Recommendations for the Collections Head, based on the supplied synthetic call data.

01 **Shift capacity to stronger windows**

Test 10:00–12:00 and 17:00–19:00 against existing allocation. The observed rate reached 33.6% at 18:00 vs ~21–22% through much of midday.

Expected signal: about 1.5× the connection rate at 18:00 vs 13:00; validate with a controlled holdout.

02 **Prioritise short, realistic promises**

Use a near-term follow-up workflow for 1–5 day PTPs; monitor kept rate by promise horizon rather than PTP count alone.

Observed kept rate: 64.9% for 1–5 days vs ~34–35% for 6–14 days.

03 **Fix BR / B7 before scaling**

Quarantine the segment, verify records, and add a wrong-number-rate alert before reintroducing it.

Observed wrong-number rate: 53.5% vs ~8.4% overall connected-call baseline; ~97 excess wrongs at this segment volume vs other segments.

Caveat: these are descriptive results from a seeded synthetic dataset; production rollout requires a live pilot and outcome tracking.